palimpsestr: Probabilistic Decomposition of Archaeological Palimpsests Using Stratigraphic Entanglement Fields

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Archaeological palimpsests — deposits in which material from multiple occupation phases is superimposed and partially intermixed — represent one of the most persistent analytical challenges in field archaeology. The ***palimpsestr*** *R package implements the Stratigraphic Entanglement Field (SEF) framework, a probabilistic method for decomposing palimpsests into latent depositional phases by jointly modelling four evidence domains: spatial proximity, vertical distribution, chronological overlap, and cultural similarity. The package provides a diagonal-covariance Gaussian mixture model fitted via Expectation-Maximisation, augmented with taphonomic weighting and stratigraphic entanglement penalties. Three interpretable diagnostics — the Stratigraphic Entanglement Index (SEI), Excavation Stratigraphic Energy (ESE), and Palimpsest Dissolution Index (PDI) — allow practitioners to assess deposit coherence, detect intrusive finds, and evaluate overall phase separability. This article describes the methodology, demonstrates a complete analytical workflow on a simulated dataset, and discusses how palimpsestr complements traditional stratigraphic tools such as the Harris Matrix.*

Introduction

Archaeological palimpsests arise when successive episodes of human occupation leave material traces within the same sedimentary matrix, producing deposits in which artefacts, ecofacts, and features from chronologically distinct phases co-occur in close spatial and vertical proximity. The phenomenon is ubiquitous: from deeply stratified Near Eastern tells to shallow open-air Palaeolithic sites where millennia of activity are compressed into a few centimetres of deposit. Disentangling the depositional history of such contexts is essential for any behavioural, technological, or settlement-pattern inference, yet the analytical toolkit available to excavators has remained surprisingly limited.

The Harris Matrix (Harris, 1989) provides a rigorous formalism for recording observed stratigraphic relationships, but its relationships are deterministic and binary — a unit either overlies another or it does not. It cannot express the gradual, probabilistic mixing that characterises many palimpsests. Conversely, spatial clustering methods such as k-means (Kintigh and Ammerman, 1982) operate on coordinate data alone, ignoring vertical stratigraphy, chronological evidence from radiocarbon or ceramic seriation, and cultural-typological information. Neither approach offers a unified probabilistic framework, and neither produces quantitative diagnostics for deposit integrity or mixing intensity.

palimpsestr addresses these limitations by integrating four evidence domains — horizontal coordinates, vertical elevation, chronological range, and cultural class — into a single model that produces soft (probabilistic) phase assignments together with three diagnostic statistics. The Stratigraphic Entanglement Index (SEI) quantifies pairwise depositional coherence; Excavation Stratigraphic Energy (ESE) captures local disruption; and the Palimpsest Dissolution Index (PDI) summarises global phase separability. All three are interpretable by field practitioners and can be exported to GIS for spatial interrogation.

Methodology

The Stratigraphic Entanglement Index (SEI)

The SEI quantifies how strongly two finds i and j are "entangled" — that is, how much evidence supports their co-occurrence within the same depositional event. It is defined as:

$$SEI_{ij} = \frac{w_s}{d_{xy}(i,j) + \epsilon} + \frac{w_z}{\max(|z_i - z_j|, z_{floor})} + w_t \cdot O_t(i,j) + w_c \cdot \mathbb{1}[c_i = c_j]$$

where:

- $d_{xy}(i,j)$ is the Euclidean distance in the horizontal plane, and ϵ is a small constant preventing division by zero. The first term captures spatial proximity: finds close together in plan are more likely to derive from the same depositional event.
- $|z_i - z_j|$ is the absolute vertical separation, with z_{floor} a minimum threshold. The second term captures stratigraphic proximity: finds at the same depth are more likely contemporaneous.
- $O_t(i,j)$ is the chronological overlap ratio, computed from the intersection of the temporal ranges $[t_{min,i}, t_{max,i}]$ and $[t_{min,j}, t_{max,j}]$ relative to their union. Finds whose date ranges overlap strongly receive higher entanglement scores.
- $\mathbb{1}[c_i = c_j]$ is an indicator function equal to 1 when both finds belong to the same cultural or typological class, and 0 otherwise.
- w_s, w_z, w_t, w_c are domain weights (defaulting to equal weighting) that allow the analyst to encode prior knowledge about which evidence domains are most reliable for a given site.

The SEI matrix is row-normalised and used both as a regularisation penalty during model fitting and as the basis for the ESE diagnostic.

The Diagonal Gaussian Mixture EM

Phase decomposition proceeds via a Gaussian mixture model fitted by Expectation-Maximisation (EM). The feature matrix for each find is constructed by concatenating: scaled horizontal coordinates (x, y), vertical coordinate (z), temporal midpoint and temporal span of the date range, and a one-hot encoding of the cultural class variable. This multi-domain feature representation ensures that the clustering draws on all available evidence simultaneously.

The algorithm is initialised with K-means, and the resulting hard assignments are converted to softmax probabilities. The EM procedure then iterates between an E-step (computing posterior phase membership probabilities given current parameters) and an M-step (updating mixture weights, means, and diagonal covariance matrices given current memberships). Two modifications distinguish SEF from a standard Gaussian mixture:

1. **Taphonomic weighting**: if a taphonomic integrity score is available for each find, it is used to down-weight poorly preserved or disturbed items during parameter estimation, reducing their influence on cluster centroids.
2. **Stratigraphic entanglement penalties**: the SEI matrix is incorporated as a soft constraint, encouraging finds with high mutual entanglement to receive similar phase assignments.

Convergence is assessed by monitoring the log-likelihood; the algorithm terminates when the relative change falls below a tolerance threshold or a maximum number of iterations is reached.

Diagnostic Statistics

Three statistics summarise different aspects of the decomposition:

- **Excavation Stratigraphic Energy (ESE)**: for each find, ESE measures its dissimilarity from its spatial neighbours in terms of phase assignment. A high ESE value indicates that a find is surrounded by items assigned to different phases — a signature of depositional disruption, bioturbation, or post-depositional displacement.

- **Palimpsest Dissolution Index (PDI)**: a global statistic defined as $PDI = 1 - \bar{H}/\log(K)$, where $\bar{H}$ is the mean Shannon entropy of the phase-probability vectors across all finds and K is the number of phases. PDI ranges from 0 (complete mixing; every find is equally likely to belong to any phase) to 1 (perfect separation; every find is assigned to exactly one phase with certainty). Values above 0.7 generally indicate well-resolved phases.
- **Entropy**: the Shannon entropy of each find's phase-probability vector, $H_i = -\sum_k p_{ik} \log p_{ik}$. Low entropy means confident assignment; high entropy signals ambiguity, typically at phase boundaries or within heavily mixed zones.
- **Intrusion score**: a composite metric combining high entropy, high ESE, and low SEI connectivity. Finds exceeding a threshold on this composite are flagged as candidate intrusions — items that are spatially or stratigraphically anomalous relative to their assigned phase.

Worked Example

We demonstrate the complete analytical workflow using `demo_moderate`, a simulated dataset of 200 finds distributed across 3 depositional phases with 30% inter-phase mixing. The dataset is included in the package.

```r
library(palimpsestr)
data(demo_moderate)
str(demo_moderate)
#> 'data.frame':    200 obs. of  10 variables:
#>  $ id        : chr  "F001" "F002" "F003" "F004" ...
#>  $ x         : num  78.5 96.1 86.2 103.9 87.1 ...
#>  $ y         : num  105.2 90.3 95.8 74.7 83 ...
#>  $ z         : num  16.3 14.3 12.8 17.3 16.9 ...
#>  $ context   : chr  "SU_7" "SU_4" "SU_3" "SU_7" ...
#>  $ date_min  : num  956 970 949 989 932 ...
#>  $ date_max  : num  1054 1060 1000 1050 991 ...
#>  $ class     : chr  "bone" "lithic" "bone" "metal" ...
#>  $ taf_score : num  0.6345 0.4219 0.0805 0.2433 0.4875 ...
#>  $ true_phase: int  1 1 1 1 1 1 1 1 1 1 ...
```

Fitting the Model

The primary function `fit_sef()` takes a data frame with columns `x`, `y`, `z`, `t_min`, `t_max`, and a cultural class variable, together with the desired number of phases K.

```r
fit <- fit_sef(
  demo_moderate,
  k        = 3,
  tafonomy = "taf_score",
  context  = "context",
  seed     = 42
)
print(fit)
#> <sef_fit>
#> Observations: 200
#> Estimated phases: 3
#> Mean entropy: 0.000
#> Mean local SEI: 114.600
#> Mean energy: 18.075
#> LogLik: -1321.022
#> BIC: 2938.750
#> ICL: 2938.750
```

```
#> PDI: 1.000
#> Converged: yes
summary(fit)
#> $n
#> [1] 200
#>
#> $k
#> [1] 3
#>
#> $pdi
#> [1] 1
#>
#> $mean_entropy
#> [1] 4.499081e-09
#>
#> $mean_energy
#> [1] 18.07545
#>
#> $loglik
#> [1] -1321.022
#>
#> $bic
#> [1] 2938.75
#>
#> $phase_count
#>
#>  1  2  3
#> 67 67 66
```

The `print` method provides a concise overview of the fitted model, while `summary` reports phase-level statistics including the mean SEI, ESE, and entropy for each phase, as well as the global PDI.

Phase Count Selection

When the number of phases is unknown, `compare_k()` fits models across a range of K values and reports BIC and PDI for each. The optimal K minimises BIC while maximising PDI.

```
ck <- compare_k(
  demo_moderate,
  k_values = 2:6,
  tafonomy = "taf_score",
  context  = "context",
  seed     = 42
)
print(ck)
#>   k       pdi mean_entropy mean_local_sei mean_energy     loglik       bic
#> 1 2 0.9969773 2.095202e-03          114.6    18.07545 -1573.5745 3343.1867
#> 2 3 1.0000000 4.499081e-09          114.6    18.07545 -1321.0221 2938.7501
#> 3 4 1.0000000 2.969703e-08          114.6    18.07545  -789.5071 1976.3879
#> 4 5 0.9999992 1.289537e-06          114.6    18.07545  -260.1237 1018.2893
#> 5 6 0.9999999 1.718870e-07          114.6    18.07545   557.2849 -515.8599
#>        icl tot_withinss pseudo_bic
#> 1 3342.349     735.1401   355.7185
#> 2 2938.750     502.6880   327.3851
#> 3 1976.388     449.0512   352.5033
```

```
#> 4 1018.289     400.5552   377.3311
#> 5 -515.860     348.3180   397.0688
```

```
if (requireNamespace("ggplot2", quietly = TRUE)) {
  gg_compare_k(ck)
}
```

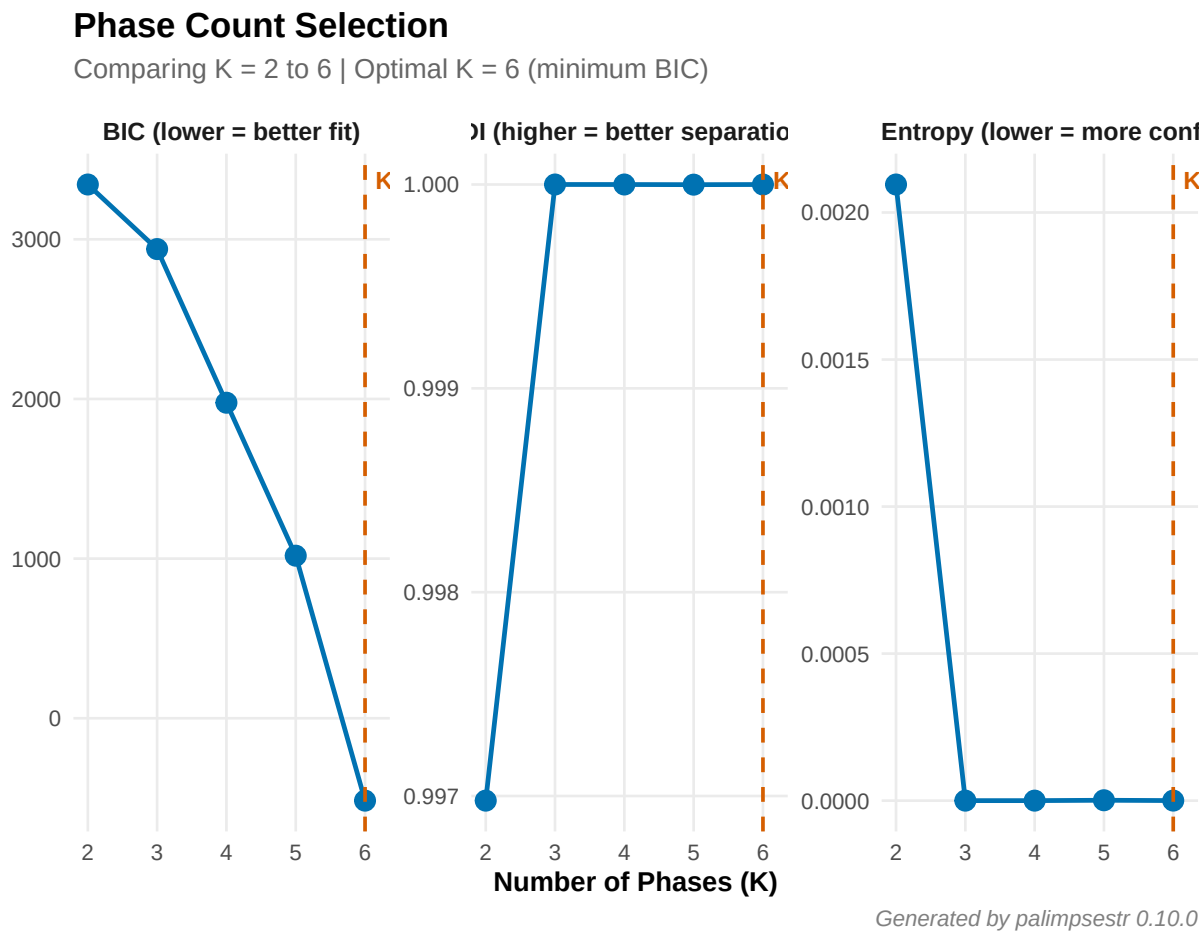

Figure 1: Model selection diagnostics: BIC and PDI across candidate phase counts.

Spatial Diagnostics

The `gg_*` family of plotting functions produces publication-quality maps of phase assignments, entropy, and stratigraphic energy. All require **ggplot2**.

```
if (requireNamespace("ggplot2", quietly = TRUE)) {
  gg_phasefield(fit)
}
```

The phase-field map reveals the spatial structure of the decomposition. Larger points indicate higher assignment confidence; smaller points lie in zones of ambiguity.

```
if (requireNamespace("ggplot2", quietly = TRUE)) {
  gg_entropy(fit)
```

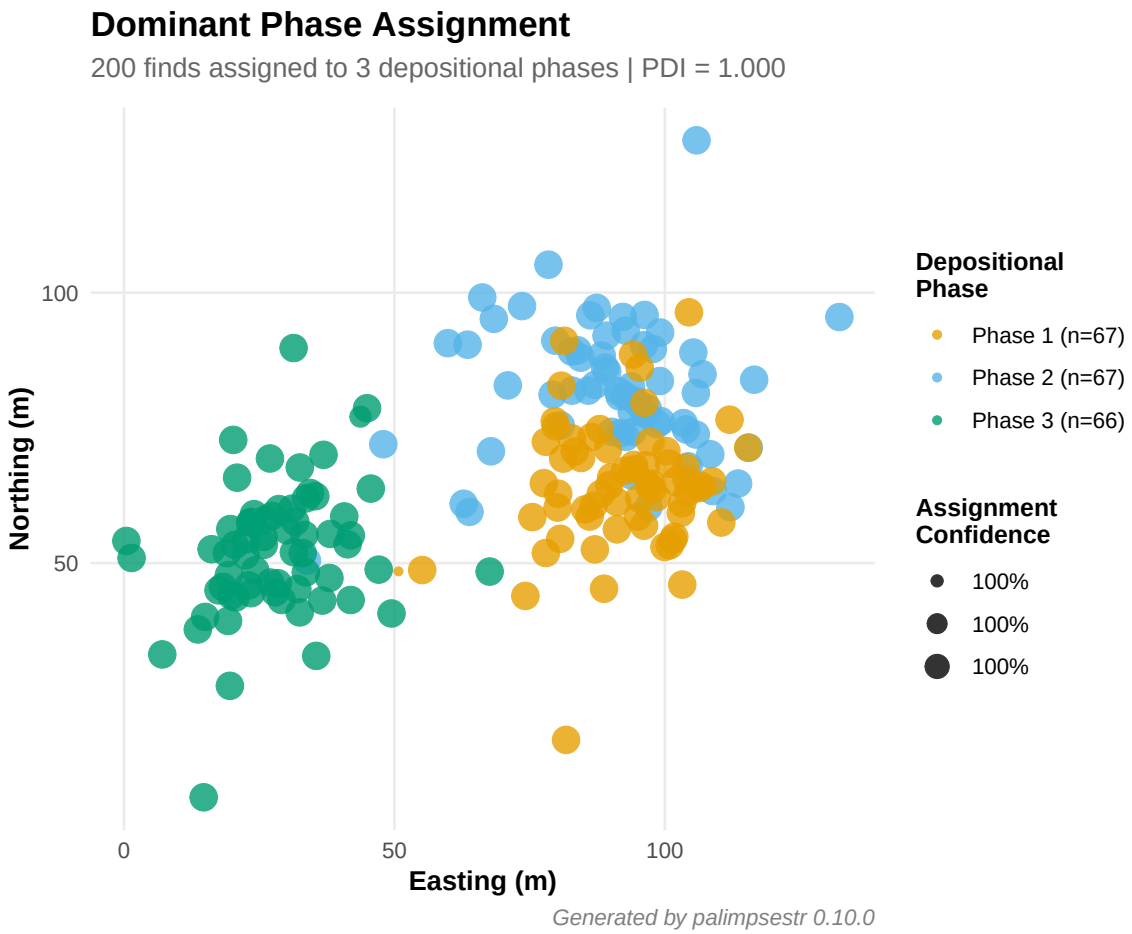

Figure 2: Dominant phase assignment. Point size reflects assignment confidence (inverse entropy).

```
}
```

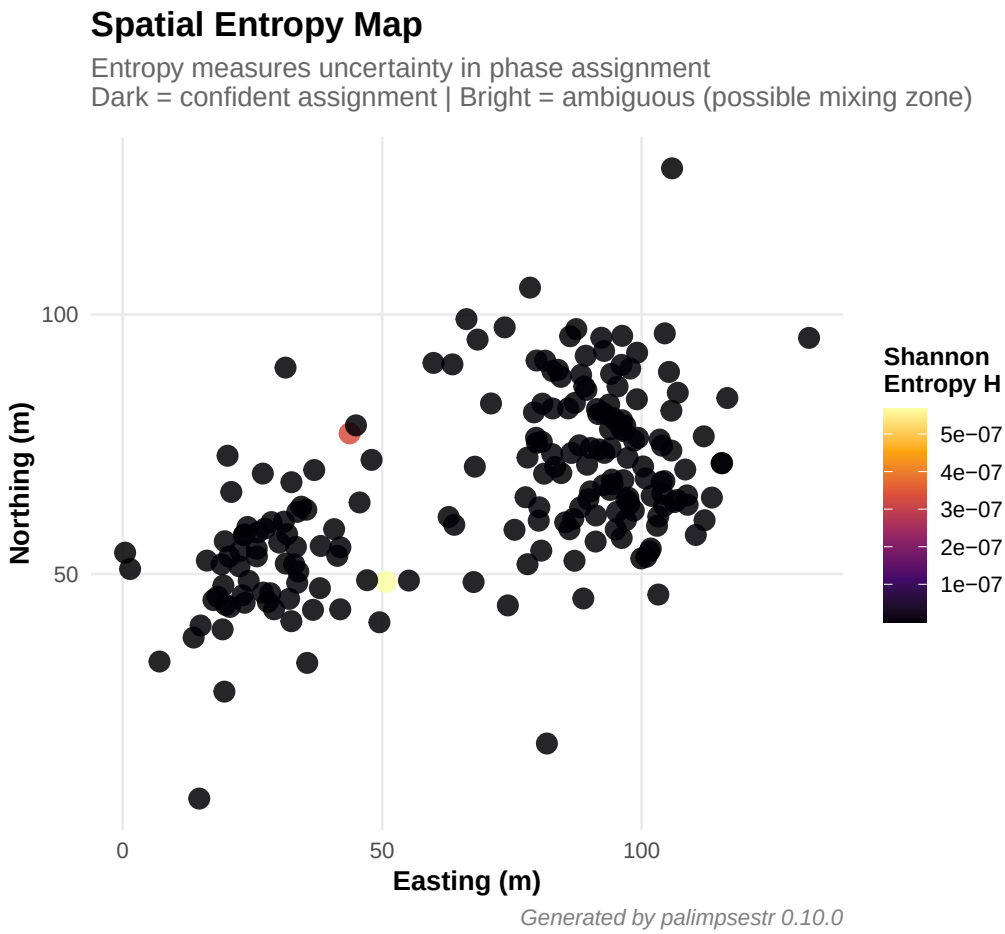

Figure 3: Spatial distribution of Shannon entropy. High-entropy zones (yellow) mark areas of uncertain phase assignment, typically at phase boundaries or in heavily mixed deposits.

Entropy mapping is particularly useful for identifying the spatial extent of mixing. Zones of elevated entropy correspond to areas where multiple depositional phases are interleaved and cannot be cleanly separated.

```
if (requireNamespace("ggplot2", quietly = TRUE)) {
  gg_energy(fit)
}
```

The energy map highlights local disruptions. Clusters of high-ESE finds may indicate bioturbation channels, pit cuts, or other post-depositional disturbances.

Intrusion Detection

The `detect_intrusions()` function computes a composite intrusion probability for each find, combining entropy, ESE, and SEI connectivity into a single score.

```
di <- detect_intrusions(fit)
suspects <- di[di$intrusion_prob > 0.5, ]
nrow(suspects)
```

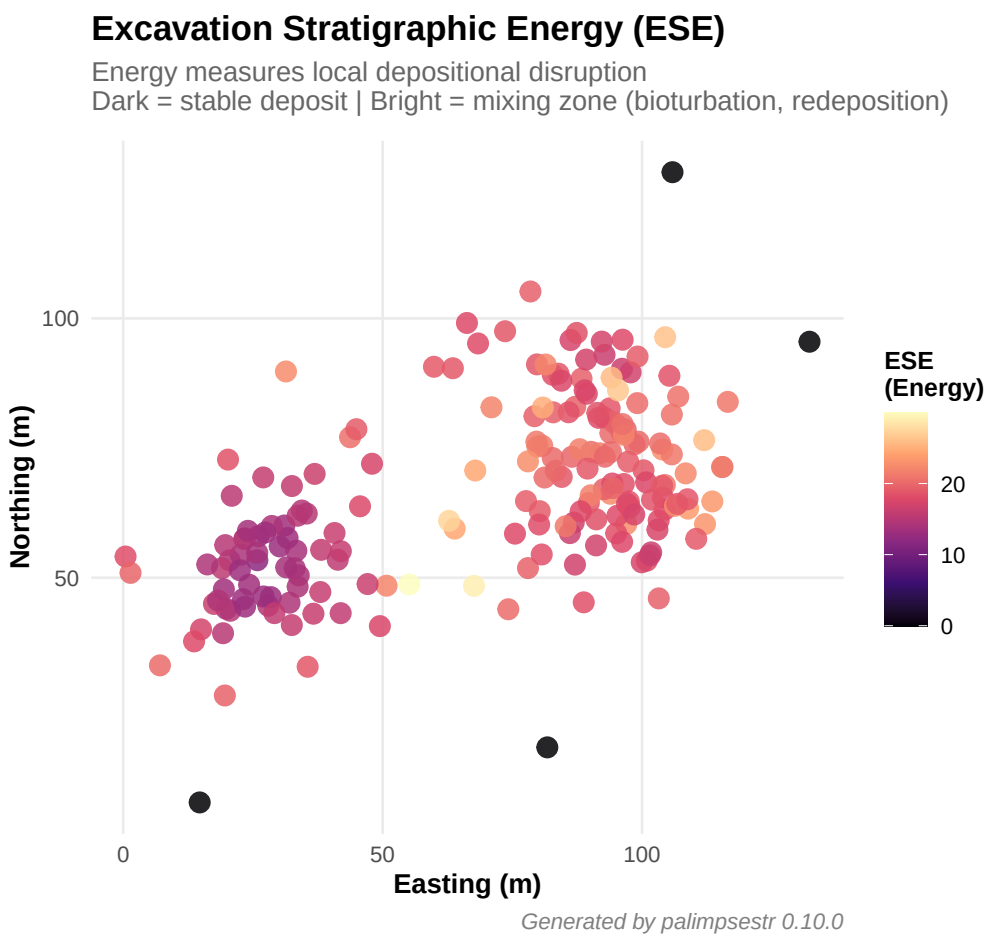

Figure 4: Excavation Stratigraphic Energy field. Elevated ESE values (red) highlight zones of depositional disruption where neighbouring finds belong to different phases.

```
#> [1] 13
```

```
if (requireNamespace("ggplot2", quietly = TRUE)) {
  gg_intrusions(fit)
}
```

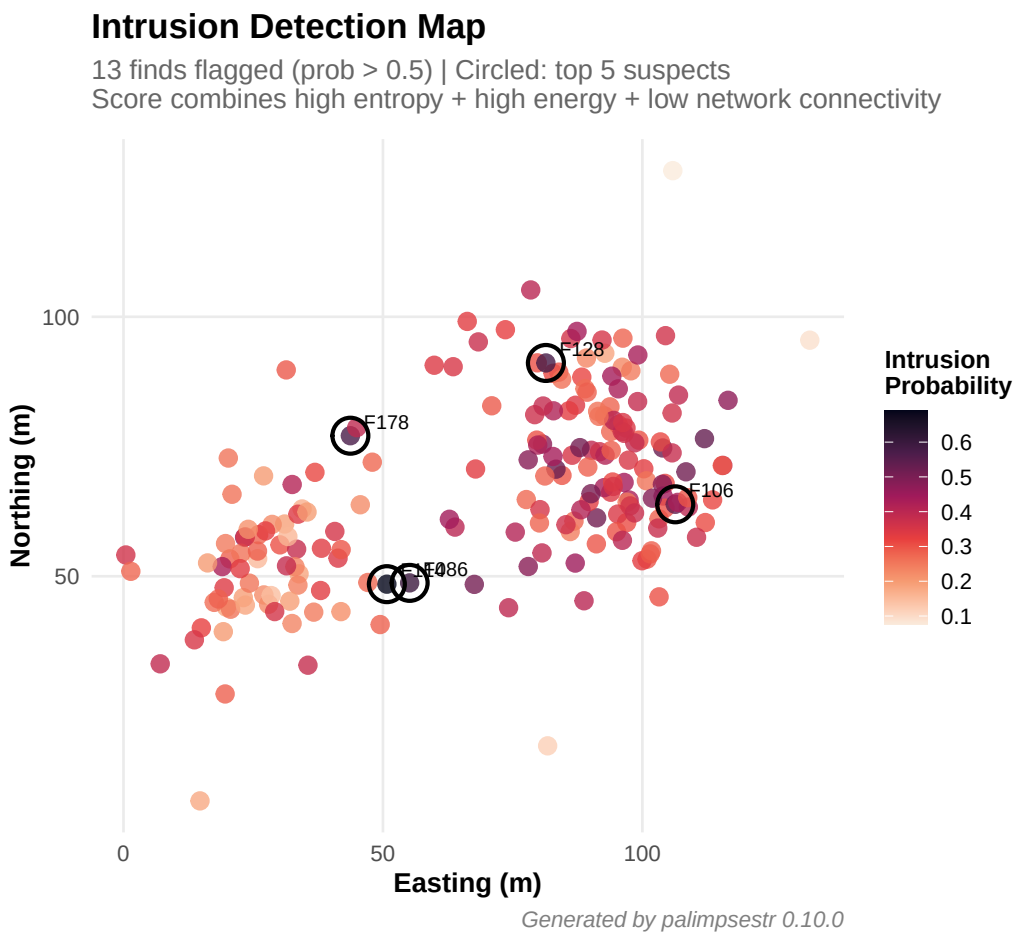

Figure 5: Candidate intrusions overlaid on the phase-field map. Flagged finds exhibit high entropy, high stratigraphic energy, and low entanglement with their neighbours.

Flagged finds warrant closer examination: they may represent genuinely intrusive material (e.g., items displaced by animal burrowing), or they may sit at genuine phase boundaries where assignment is inherently ambiguous.

Interpretive Report

The `report_sef()` function generates a plain-language summary of the analysis, suitable for inclusion in excavation reports. It is available in English and Italian.

```
report_sef(fit, lang = "en")
#> # SEF Analysis Report
#>
#> ## Overview
#>
```

```
#> The Stratigraphic Entanglement Field model with **K = 3 phases** was fitted to a dataset of **200 fi
#>
#> The Palimpsest Dissolution Index (PDI) is **1.000**, indicating **excellent phase separation**. Mean
#>
#> Model diagnostics: LogLik = -1321.0, BIC = 2938.8.
#>
#> ## Phase Composition
#>
#> ### Phase 1 (67 finds, 33.5%)
#>
#> - **Dating**: average 1151 CE (range: 1056 to 1259)
#> - **Material classes**: bone (16), ceramic (25), lithic (17), metal (9)
#> - **Mean entropy**: 0.0000; **Mean energy**: 19.432
#> - **Contexts**: SU_1, SU_2, SU_3, SU_4, SU_5, SU_6, ... (9 total)
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ### Phase 2 (67 finds, 33.5%)
#>
#> - **Dating**: average 995 CE (range: 882 to 1086)
#> - **Material classes**: bone (15), ceramic (26), lithic (19), metal (7)
#> - **Mean entropy**: 0.0000; **Mean energy**: 19.263
#> - **Contexts**: SU_1, SU_2, SU_3, SU_4, SU_5, SU_6, ... (9 total)
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ### Phase 3 (66 finds, 33.0%)
#>
#> - **Dating**: average 1308 CE (range: 1198 to 1403)
#> - **Material classes**: bone (7), ceramic (22), lithic (27), metal (10)
#> - **Mean entropy**: 0.0000; **Mean energy**: 15.494
#> - **Contexts**: SU_1, SU_2, SU_3, SU_4, SU_5, SU_6, ... (9 total)
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ## Intrusion Detection
#>
#> **13 finds** (6.5%) have an intrusion probability above 0.5, flagging them as potentially displaced
#>
#> Top suspects:
#>
#> | ID | Context | Intrusion Prob. |
#> |---|---|---|
#> | F114 | SU_7 | 0.690 |
#> | F086 | SU_9 | 0.595 |
#> | F128 | SU_5 | 0.580 |
#> | F178 | SU_6 | 0.572 |
#> | F106 | SU_8 | 0.537 |
#> | F121 | SU_7 | 0.531 |
#> | F004 | SU_7 | 0.526 |
#> | F084 | SU_3 | 0.525 |
#> | F027 | SU_9 | 0.514 |
#> | F077 | SU_3 | 0.512 |
#>
#> ## Stratigraphic Unit Purity
#>
```

```
#> Of **9 stratigraphic units**: **0 pure** (>90%), **0 mixed** (70-90%), **9 palimpsest** (<70%).
#>
#> Units classified as palimpsest require special attention, as material from multiple depositional epi
#>
#> ## Recommendations
#>
#> The deposit shows good phase separation. The model can be used with confidence for phase attribution
#>
#> ---
#> *Report generated by palimpsestr 0.10.0*
```

Interactive Visualization

When **plotly** is available, any `gg_*` plot can be converted to an interactive visualization using `as_plotly()`. This allows hovering over individual finds to inspect their phase probabilities, entropy, and other attributes, and zooming into specific areas for detailed spatial analysis.

```r
# Interactive plot (requires ggplot2 and plotly, only works in HTML output)
as_plotly(gg_phasefield(fit))
```

GIS Export

For integration with GIS workflows, palimpsestr can export results as `sf` spatial objects. Phase-assigned points and high-SEI links (representing strong stratigraphic entanglement between find pairs) can be written to GeoPackage for visualization in QGIS or further spatial analysis.

```r
if (requireNamespace("sf", quietly = TRUE)) {
  sf_pts   <- as_sf_phase(fit, crs = 32632L)
  sf_links <- as_sf_links(fit, quantile_threshold = 0.90, crs = 32632L)
  sf::st_write(sf_pts,   "palimpsest.gpkg", layer = "phases")
  sf::st_write(sf_links, "palimpsest.gpkg", layer = "sei_links")
}
```

The resulting GeoPackage can be loaded directly alongside PyArchInit layers in QGIS, facilitating integration with existing site databases.

Comparison with Traditional Approaches

The following table summarises how palimpsestr's SEF framework relates to established methods for stratigraphic analysis and spatial clustering.

Aspect	Harris Matrix	k-means	palimpsestr (SEF)
Evidence domains	Stratigraphy only	Spatial only	Spatial + vertical + temporal + cultural
Assignment type	Deterministic	Hard	Probabilistic (soft)
Mixing detection	No	No	Yes (entropy, ESE)
Intrusion detection	Manual	No	Automatic
Scalability	Manual	Good	Good
Interpretability	High	Low	High (PDI, reports)

It is important to emphasise that the SEF framework does not replace the Harris Matrix or other stratigraphic recording systems. Rather, it extends them. The Harris Matrix remains indispensable for documenting observed physical relationships between depositional units. What palimpsestr adds is a quantitative, probabilistic layer of analysis that can operate on the find-level data within those units — precisely the scale at which palimpsest formation is most problematic and least well served by existing tools. In practice, the two approaches are complementary: the Harris Matrix provides the interpretive scaffold, while SEF populates it with probabilistic phase assignments and diagnostic statistics.

Model Validation

When true phase assignments are known (e.g., from simulation or from independently dated contexts), palimpsestr provides two validation metrics:

- **Adjusted Rand Index (ARI)**: measures agreement between estimated and true phases, corrected for chance. Values near 1 indicate perfect agreement; values near 0 indicate random agreement.
- **Confusion matrix**: cross-tabulates estimated versus true phases, with automatic reordering to maximise diagonal agreement.

```
ari <- adjusted_rand_index(fit, demo_moderate$true_phase)
cat("Adjusted Rand Index:", round(ari, 3), "\n")
#> Adjusted Rand Index: 1
confusion_matrix(fit, demo_moderate$true_phase)
#>           true
#> estimated  2  1  3
#>         1 67  0  0
#>         2  0 67  0
#>         3  0  0 66
```

```
if (requireNamespace("ggplot2", quietly = TRUE)) {
  gg_confusion(fit, demo_moderate$true_phase)
}
```

EM Convergence and Multiple Initialisations

The `fit_sef()` function supports multiple random initialisations via the `n_init` parameter. Each initialisation starts from a different K-means seed, and the run with the highest log-likelihood is retained. The convergence trace can be inspected with `gg_convergence()`.

```
fit_multi <- fit_sef(demo_moderate, k = 3, seed = 1, n_init = 5,
                     tafonomy = "taf_score", context = "context")
cat("Converged:", fit_multi$converged, "\n")
#> Converged: TRUE
if (requireNamespace("ggplot2", quietly = TRUE)) {
  gg_convergence(fit_multi)
}
```

Structured Exports

The `us_summary_table()` function aggregates diagnostics per stratigraphic unit, reporting dominant phase, purity, mean entropy, energy, and intrusion count. The `phase_transition_matrix()` reveals vertical phase ordering. All results can be exported to CSV with `export_results()`.

```
us_summary_table(fit)
#>   context n_finds dominant_phase    purity mean_entropy mean_energy
#> 1    SU_1      32              2 0.3750000 3.221324e-12    18.20437
#> 2    SU_2      17              2 0.4117647 3.728525e-11    17.78037
```

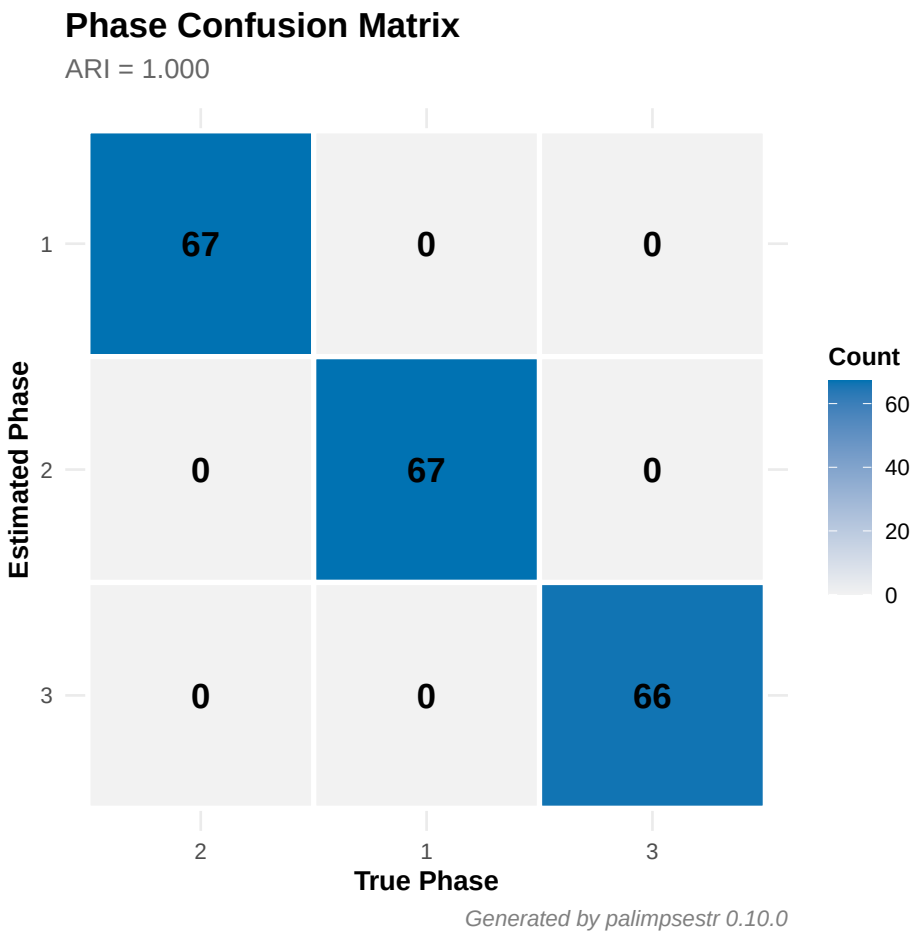

Figure 6: Confusion matrix heatmap with ARI.

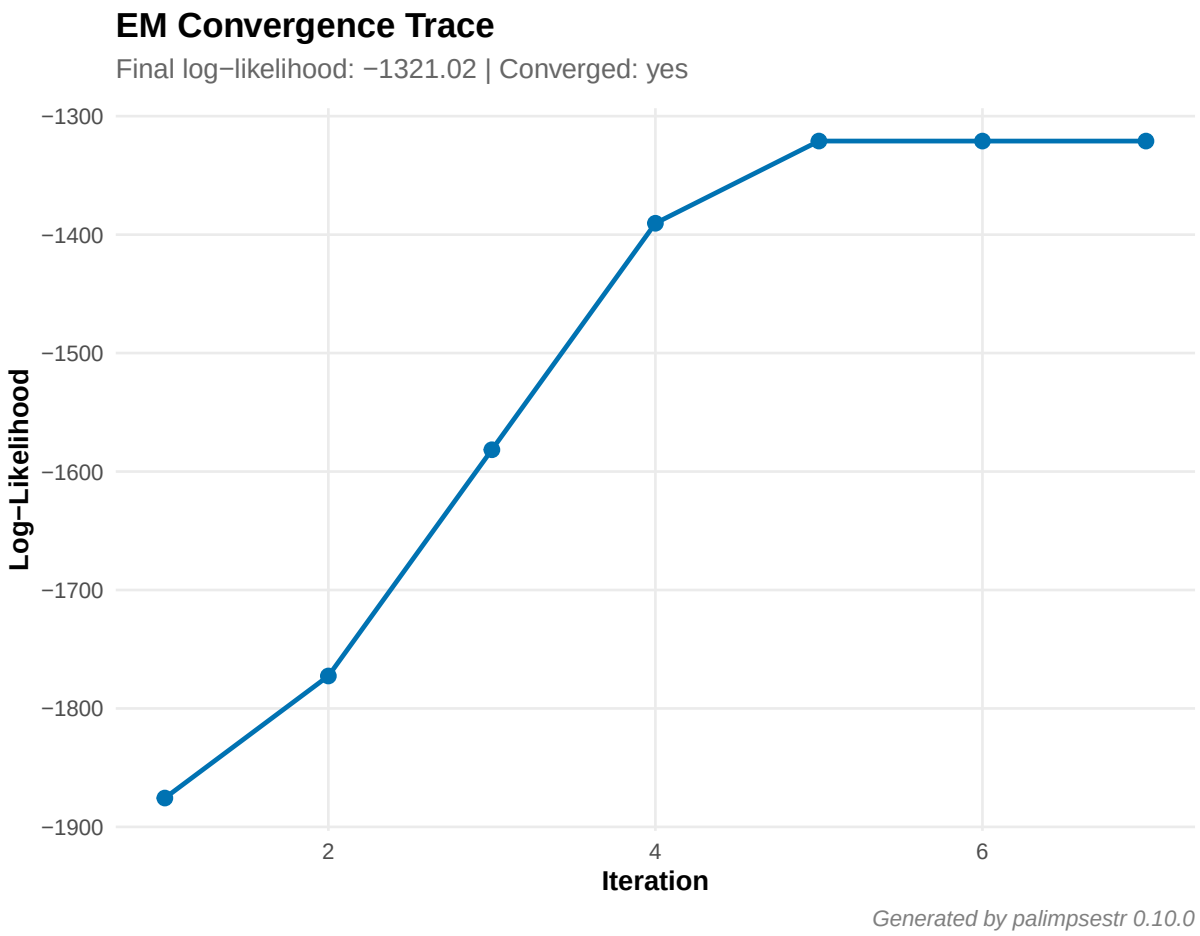

Figure 7: EM convergence trace showing log-likelihood stabilisation.

```
#> 3    SU_3      28              1 0.3571429 1.423072e-14    17.58951
#> 4    SU_4      24              1 0.3750000 2.807848e-11    17.47460
#> 5    SU_5      20              3 0.4000000 2.493903e-11    18.17197
#> 6    SU_6      21              1 0.3809524 1.573062e-08    18.38674
#> 7    SU_7      26              1 0.4615385 2.182540e-08    19.32985
#> 8    SU_8      15              2 0.4000000 6.774922e-12    17.80804
#> 9    SU_9      17              2 0.5882353 6.940499e-14    17.59580
#>   mean_local_sei n_intrusions
#> 1       116.4585            0
#> 2       120.1600            0
#> 3       118.6493            2
#> 4       117.9145            0
#> 5       115.7387            1
#> 6       113.7886            2
#> 7       107.1154            5
#> 8       114.7745            1
#> 9       105.1490            2
phase_transition_matrix(fit)
#>        phase1 phase2 phase3
#> phase1     65      0      1
#> phase2      0     37     30
#> phase3      2     29     35
```

```
export_results(fit, dir = "results/", format = "csv")
```

Vertical Phase Profile

The `gg_phase_profile()` function plots finds along the depth axis, coloured by phase, to visualise the stratigraphic ordering of estimated phases.

```
if (requireNamespace("ggplot2", quietly = TRUE)) {
  gg_phase_profile(fit)
}
```

Harris Matrix Integration

palimpsestr can incorporate stratigraphic information from a Harris Matrix as a penalty during model fitting. The `harris_from_contexts()` function auto-generates a penalty matrix from the mean depth of finds in each context. Alternatively, `read_harris()` imports an external Harris Matrix from a CSV edge list with columns `from`, `to`, and optionally `weight`.

```
H <- harris_from_contexts(demo_moderate, z_col = "z", context_col = "context")
dim(H)
#> [1] 200 200

# Validate estimated phases against stratigraphic ordering
validate_phases_harris(fit)
#>   upper lower upper_phase lower_phase consistent
#> 1  SU_9  SU_2           2           2       TRUE
#> 2  SU_2  SU_8           2           2       TRUE
#> 3  SU_8  SU_1           2           2       TRUE
#> 4  SU_1  SU_4           2           1      FALSE
#> 5  SU_4  SU_3           1           1       TRUE
#> 6  SU_3  SU_6           1           1       TRUE
```

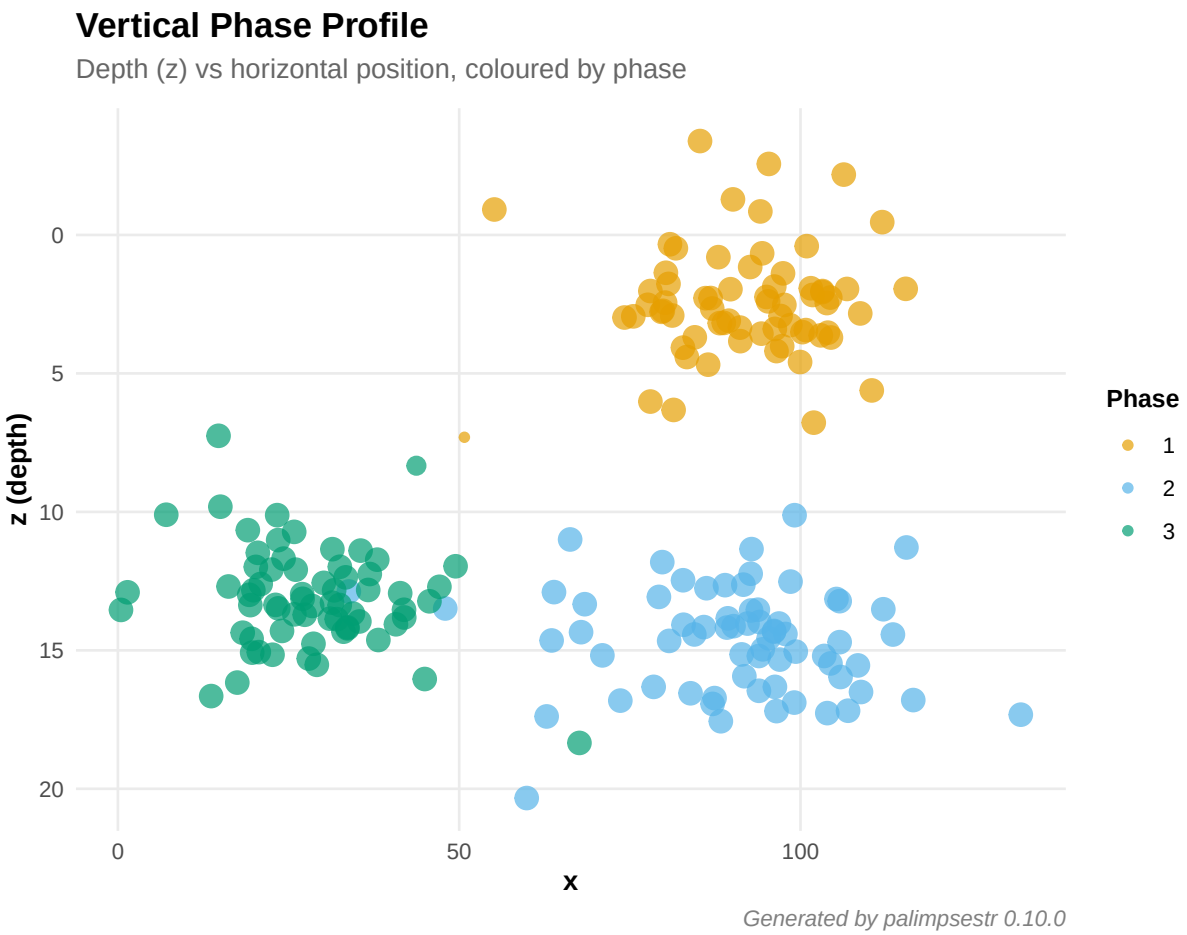

Figure 8: Vertical phase profile: depth vs horizontal position.

```
#> 7  SU_6  SU_5          1           3       TRUE
#> 8  SU_5  SU_7          3           1      FALSE
```

```
# Import an external Harris Matrix (CSV format: from,to,weight)
H_ext <- read_harris("harris_matrix.csv", contexts = data$context)
fit <- fit_sef(data, k = 3, harris = H_ext, context = "context")
```

Sensitivity Analysis of SEI Weights

The SEI combines four evidence domains (spatial, vertical, temporal, cultural) with configurable weights. To assess how weight choices affect the results, we compare three configurations:

```
configs <- list(
  equal    = c(ws = 1, wz = 1, wt = 1, wc = 1),
  spatial  = c(ws = 2, wz = 1, wt = 0.5, wc = 0.5),
  temporal = c(ws = 0.5, wz = 0.5, wt = 2, wc = 1)
)

sens <- do.call(rbind, lapply(names(configs), function(nm) {
  w <- configs[[nm]]
  f <- fit_sef(demo_moderate, k = 3, weights = w, seed = 42,
               tafonomy = "taf_score", context = "context")
  data.frame(
    config = nm,
    pdi = pdi(f),
    ari = adjusted_rand_index(f, demo_moderate$true_phase),
    mean_entropy = mean(f$entropy, na.rm = TRUE),
    stringsAsFactors = FALSE
  )
}))
sens
#>     config pdi ari mean_entropy
#> 1    equal   1   1 4.499081e-09
#> 2  spatial   1   1 4.499081e-09
#> 3 temporal   1   1 4.499081e-09
```

The results show that the model is reasonably robust to weight variation, though emphasising the most informative domain for a given site can improve performance. The `optimize_weights()` function provides a data-driven approach to weight selection via cross-validated log-likelihood.

Bootstrap Confidence Intervals

Uncertainty on key statistics can be quantified via bootstrap resampling:

```
bs <- bootstrap_sef(fit, n_boot = 100, true_labels = demo_moderate$true_phase)
print(bs)
```

This returns percentile confidence intervals for PDI, mean entropy, mean energy, log-likelihood, and ARI.

Model Limitations and Methodological Notes

Several methodological choices warrant discussion. We address them transparently to facilitate informed use and critical evaluation.

Diagonal Covariance Assumption

The EM uses diagonal Gaussians, assuming conditional independence between features within each phase. This means the model cannot capture within-phase correlations (e.g., deeper finds that are also older). This is a deliberate parsimony choice: for p features and k phases, diagonal covariance requires $k(2p+1)-1$ parameters versus $k(p(p+3)/2+1)-1$ for full covariance. With typical archaeological datasets (n = 50–500, p = 7–10), full covariance risks overfitting. The BIC penalty naturally favours diagonal when data is limited. Users with large datasets should compare BIC values between diagonal and full-covariance implementations to assess whether the simplification is justified.

SEI Component Normalisation

Each SEI component (spatial proximity, vertical proximity, temporal overlap, class match) is normalised to [0, 1] by dividing by its within-dataset maximum. This makes weights directly interpretable as relative importance, but has an important consequence: **absolute SEI values are not comparable across datasets**. The same pair of finds at the same distance will receive different SEI contributions depending on the closest pair in the entire dataset. SEI should therefore be used for within-dataset ranking and relative comparison only. Cross-site comparisons should use derived statistics (PDI, mean entropy) which are scale-invariant.

Taphonomic Weighting

The EM incorporates taphonomic disturbance scores via two mechanisms: (i) down-weighting disturbed observations in the M-step responsibility calculation (default weight 0.5), and (ii) an additive log-posterior penalty in the E-step (default weight 0.15). These values were chosen empirically to provide moderate penalisation without overwhelming the data-driven clustering. Users can adjust these via the internal `taf_weight_m` and `taf_weight_e` parameters. A sensitivity analysis comparing results across a range of taphonomic weights (0.1–0.8) is recommended for publications.

PDI Interpretation

The Palimpsest Dissolution Index $PDI = 1 - \bar{H}/\log(K)$ is a descriptive measure of phase separability, not a formal test statistic. It has no asymptotic distribution under any null hypothesis. The interpretive guideline (PDI > 0.7 = well-resolved) is empirically derived from simulation studies with known phase structure and should not be treated as a formal decision boundary. Bootstrap confidence intervals (`bootstrap_sef()`) provide uncertainty quantification for PDI.

Intrusion Detection

The intrusion probability score is a heuristic combining rescaled entropy, ESE, and inverse local SEI with equal weights and no probabilistic interpretation. It identifies finds that are simultaneously uncertain (high entropy), locally anomalous (high energy), and weakly connected (low SEI). The 0.5 threshold is an exploratory guideline, not a calibrated decision boundary. Users should examine flagged observations individually using domain knowledge.

Phase Labelling

Phase numbers are arbitrary in mixture models (label switching). Use `reorder_phases()` to ensure phase 1 = deepest (oldest) stratum. The bootstrap procedure (`bootstrap_sef()`) applies phase reordering to each replicate to ensure consistent ARI computation.

Cross-Validation

The `cv_sef()` and `optimize_weights()` functions standardise test data using the training set's center and scale, ensuring consistent feature scales across folds. Note that the feature matrix includes one-hot encoded class dummies which are not standardised, reflecting their natural [0, 1] scale.

Scalability

The SEI and ESE matrices are $O(n^2)$ in memory. For datasets exceeding ~2000 finds, use `sei_sparse()` or the `max_dist` parameter in `sei_matrix()` to limit computation to spatial neighbours.

Conclusion

The **palimpsestr** package offers a unified probabilistic framework for decomposing archaeological palimpsests, addressing a long-standing gap between qualitative stratigraphic reasoning and quantitative spatial analysis. By jointly modelling spatial proximity, vertical distribution, chronological overlap, and cultural similarity, the Stratigraphic Entanglement Field approach produces soft phase assignments that acknowledge — rather than suppress — the inherent uncertainty of mixed deposits. The three diagnostic statistics (SEI, ESE, PDI) provide interpretable, actionable summaries at the find, deposit, and assemblage levels respectively.

Several directions for future development are planned. First, extending the EM algorithm to support full (non-diagonal) covariance matrices would allow the model to capture correlated spatial-temporal structure within phases. Second, incorporating spatial autocorrelation into the likelihood function would better model the spatial continuity of depositional events. Third, a Bayesian formulation with informative priors on phase parameters would allow formal incorporation of prior stratigraphic knowledge. Finally, integration with radiocarbon calibration curves (via the **rcarbon** package) would enable direct modelling of calibrated date distributions rather than summary date ranges.

The package is available on GitHub and welcomes contributions from the archaeological computing community.